



RESmart™ CPAP System

# **Service Manual**

# Important

All data in this manual have been verified correctly. It is sufficient for servicing and repairing this device. If applying this manual on other purpose, the validation should be confirmed by BMC Medical Co. Other wise, BMC Medical Co. has no responsibility for the result. All information in this manual is protected by law and regulation. All copyrights are belongs to BMC Medical Co.

INPORTANT, CAUTION AND WARNING in this manual are to emphasize dangers to service people.

**WARNING**      If do not operate properly, may cause damage to people and environment.

**CAUTION**      If do not operate properly, may cause damage to instrument.

**IMPORTANT**    Important information for servicing and repairing.

## Safety Notice

### Electric Shock

There is above 100VAC voltage inside this device, please be sure to repair device after power off.

### Chemical Safety

There may be risk of virus after touching by patient. Please clean device or wear protective glove before servicing and repairing. Please deal with the waste according to regulation. Wash hands by disinfectant after operating this device.

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## 1. Introduction

### 1.1 Intended Use

The RESmart system is a CPAP (Continuous Positive Airway Pressure) device designed for the treatment of adult Obstructive Sleep Apnea (OSA) only.

The RESmart is to be used only on the instruction of a licensed health care professional. Your home care provider will make the correct pressure settings according to your health care professional's prescription.

This device is not intended for life support.

This service manual is used to help service people maintenance RESmart™ CPAP system of BMC Medical Co. more efficiently. Instructions in this manual may help device work best.

Engineer can find needed instruction from this manual quickly.

**IMPORTANT** Read and understand instruction in this manual before operating.

**IMPORTANT** RESmart™ CPAP system can not be used for life support.

### 1.2 Implement

#### 1. Implement for repairing

Type '+' screw driver

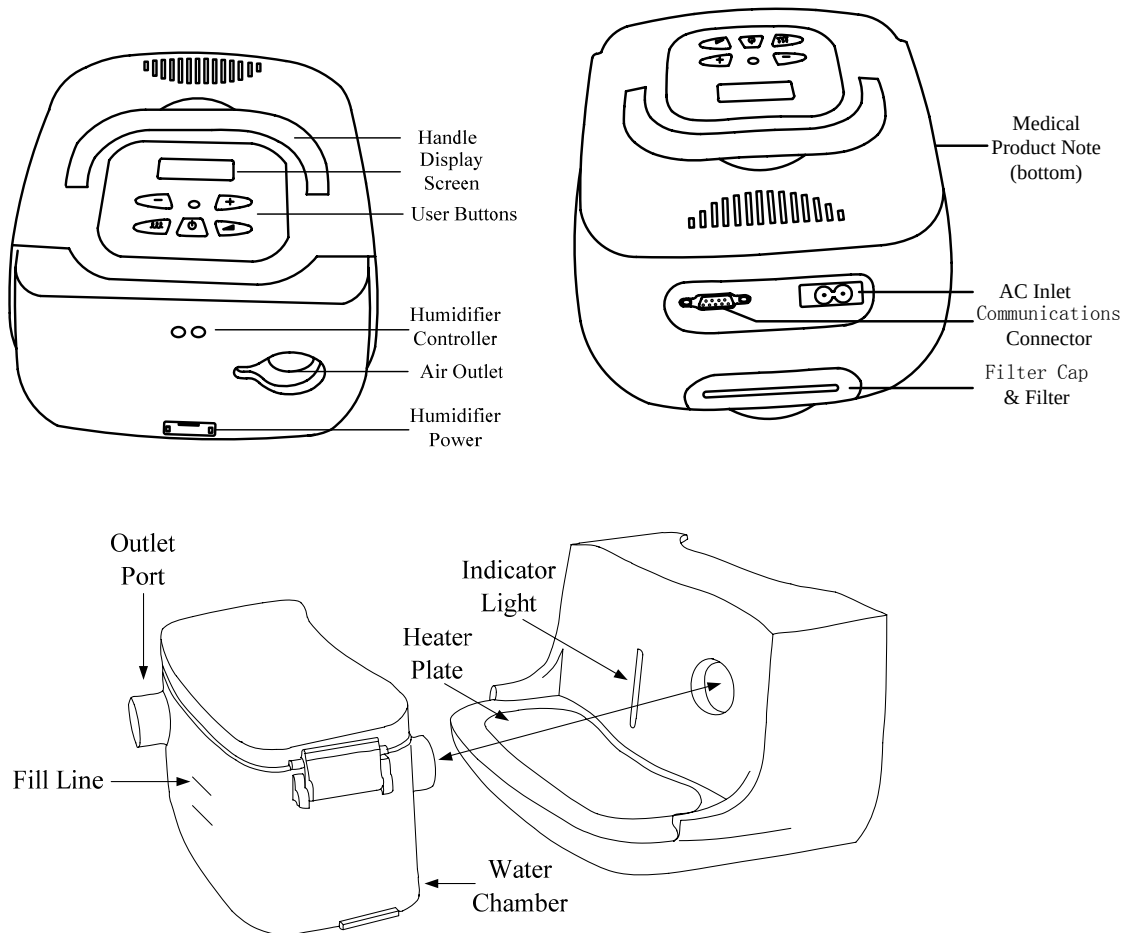
Pincers

#### 2. Implement for measuring

Pressure meter

## 2. Structure

### 2.1 Outer Illumination



### 2.2 Inner Construction

See appendix A

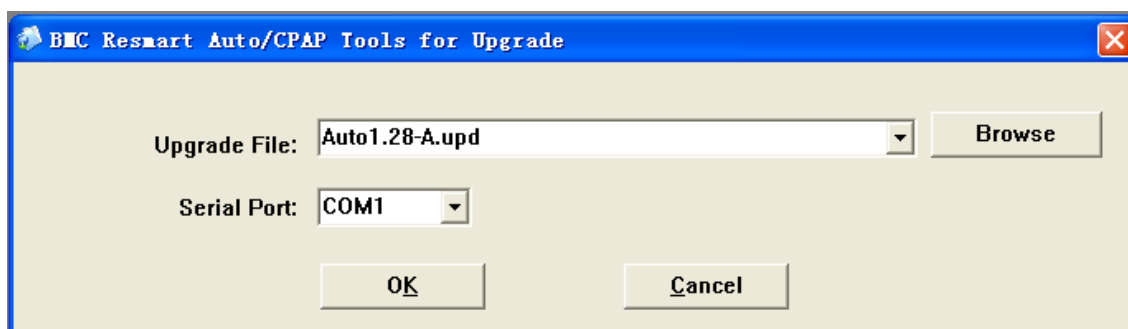
### 2.3 Spare Part

See appendix B

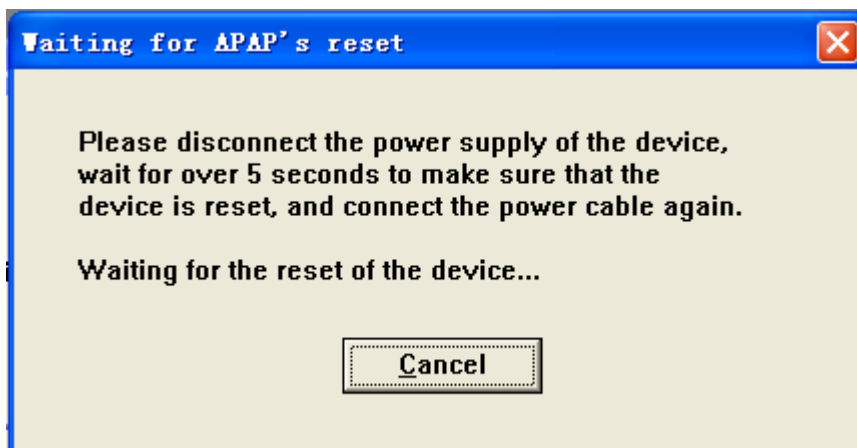
### 3. Software

#### 3.1 Instruction for Upgrading Device Software

1. Power off the RESmart CPAP/AutoCPAP device, then connect PC and the device via data cable.
2. Run the software “BMC RESmart Tools for Upgrade” from PC and display as below:

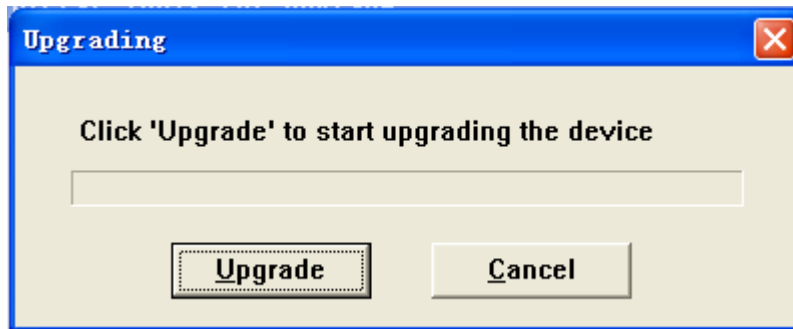


- a) Select the right Upgrade File (for example: version 1.28 for AutoCPAP is “Auto1.28-A.upd”).
  - b) Select the right serial port, default is COM1.
3. Click “OK”, display as below:



Power on the CPAP/AutoCPAP device. If the device is power on from beginning, please power off the device and turn on again after 5 seconds more.

4. After the device is turned on, the following will display:



Click “Upgrade” to start. If success, the right version number will be displayed on CPAP/AutoCPAP device screen when every power on.

**Note:** If failed during upgrading, please repeat from above 1-4 steps.

If “Error 08” displayed after software upgrading, it means unmatched software used (for example: apply AutoCPAP software to CPAP device). In this case, apply correct software and upgrade again.

## 3.2 Hidden Function

### 1. Humidifier matching and dis-matching

(Available on version 1.19 and later)

Assemble humidifier on main device, do not apply water tank. Enter maintenance menu, set ‘Treat P’=3 and ‘Ramp’=40, then press ‘On/Off’ button when ‘Manual P’ appears. Thus the humidifier is matched. When ‘OK’ appears on LCD, humidifier matching successes.

If cut the power supply after pressing ‘On/Off’ button when ‘Manual P’ appears, then the humidifier is dis-matched.

### 2. Reset

(Available on version 1.29 and later)

Enter maintenance menu, press ‘Ramp’ button for 5 seconds more, thus hidden date setting appears. Set ‘YYYY/MM/DD’=‘2097/07/01’, press ‘On/Off’ when ‘Hour’ setting, thus all device setting and use time, use days are reset.

### 3. Pressure calibration

(Available on version 1.20 and later)

Connect pressure meter to the gas outlet in the device, enter hidden date setting, set ‘YYYY/MM/DD’=‘2098/08/08’, press ‘On/Off’ when ‘Hour’ setting, thus pressure calibration starts. The

device increases pressure to 20hPa(cmH<sub>2</sub>O), AD sampling data will appear on LCD. When LCD data displaying is stable, if pressure meter value is not 20, press '+/-' button to adjust the pressure output. Until output pressure value is 20 and LCD displays stable, press 'On/Off' button to finish pressure calibration.

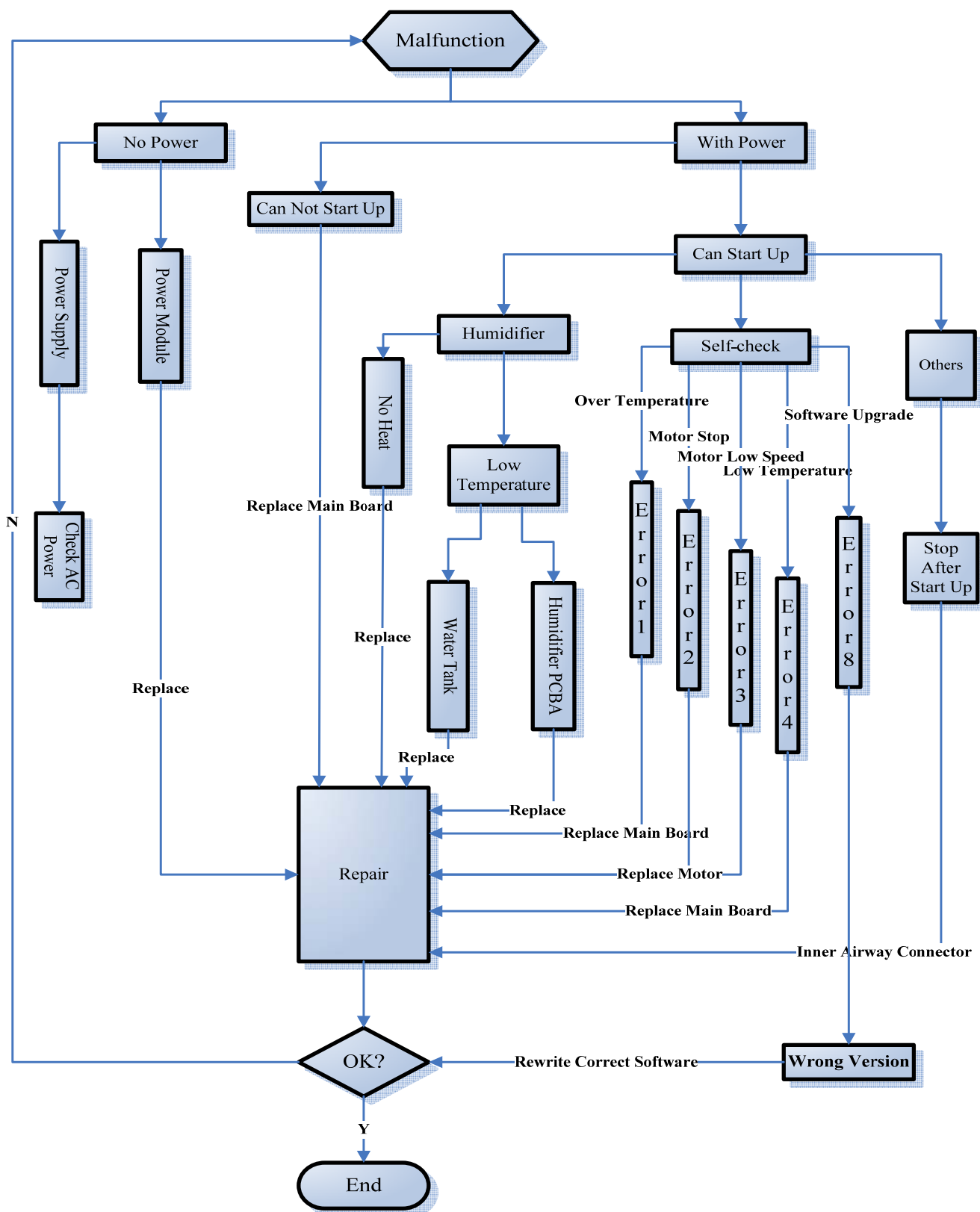
#### **4. Debug mode**

(Available on version 1.19 and later)

Enter hidden date setting, set 'YYYY/MM/DD'='2099/08/02(or 03)', press 'On/Off' when 'Hour' setting, thus debug #02 or #03 mode starts. The device can send real time data to computer via serial port. Only cutting off power supply can make device back to normal mode.

## 4. Malfunction and Countermeasure

### 4.1 Malfunction Determination



## 4.2 Error List and Information

| Error Index | Problem                 | Causing                            |
|-------------|-------------------------|------------------------------------|
| Error 1     | Over temperature inside | Malfunction on main board          |
| Error 2     | Motor stop              | Malfunction on motor or controller |
| Error 3     | Motor low speed         | Malfunction on motor               |
| Error 4     | Low temperature inside  | Malfunction on main board          |
| Error 8     | Wrong software version  | Software                           |

## Appendices

### Appendix A:

#### Device Configuration

##### 1. Main device

Fig. A - Main board (backside)

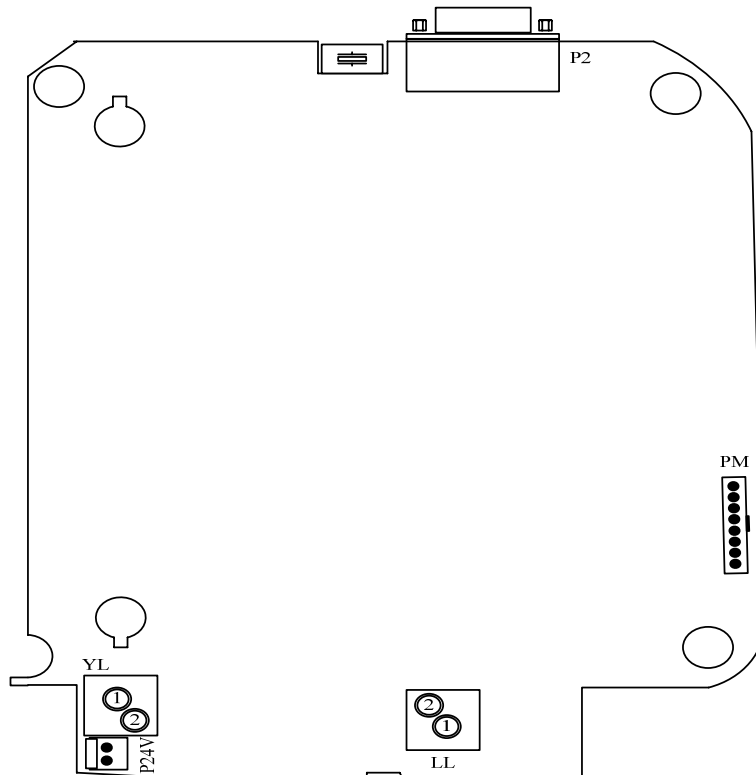
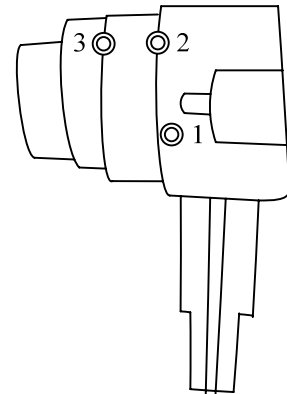


Fig. B – Outlet (leftside)



Hole #1 in Fig. B, connect to sensor YL via silicon rubber canal. Specified in Table 1:

| Fig. B | Fig. A               | Specification of canal        |
|--------|----------------------|-------------------------------|
| 1      | Hole #1 on sensor YL | $\phi$ 2mm, $\Phi$ 4mm, L12cm |
|        |                      |                               |
|        |                      |                               |

**Table 1**

Fig. C – Power PCBA (front side)

Sockets figuration

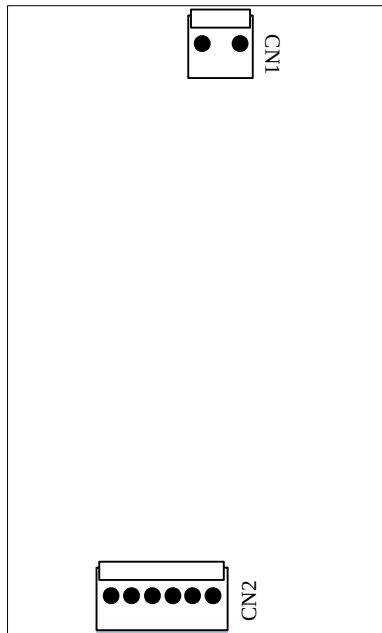


Fig. D – Wire connecting on power PCBA

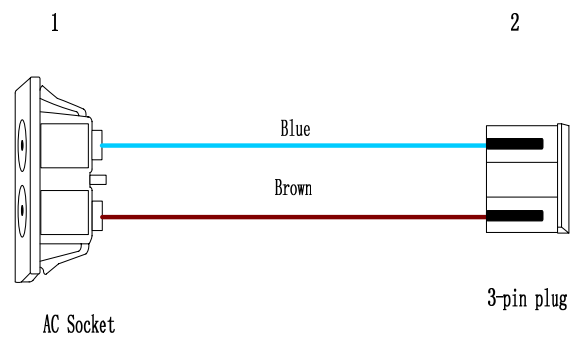


Fig. E – Power supply connecting

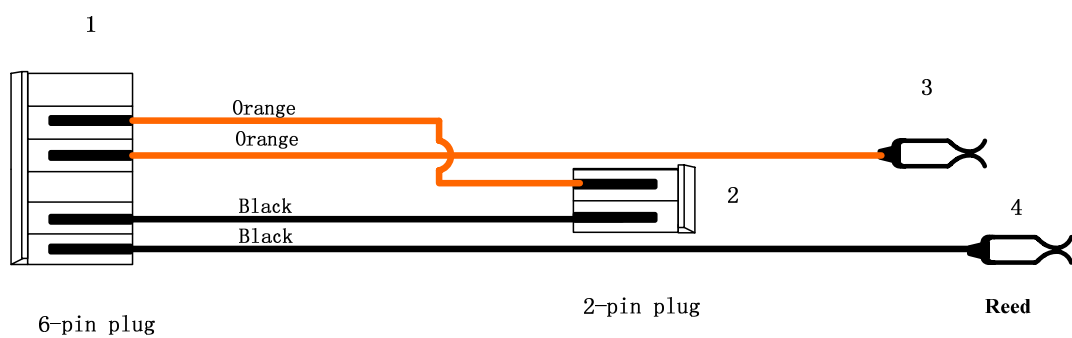
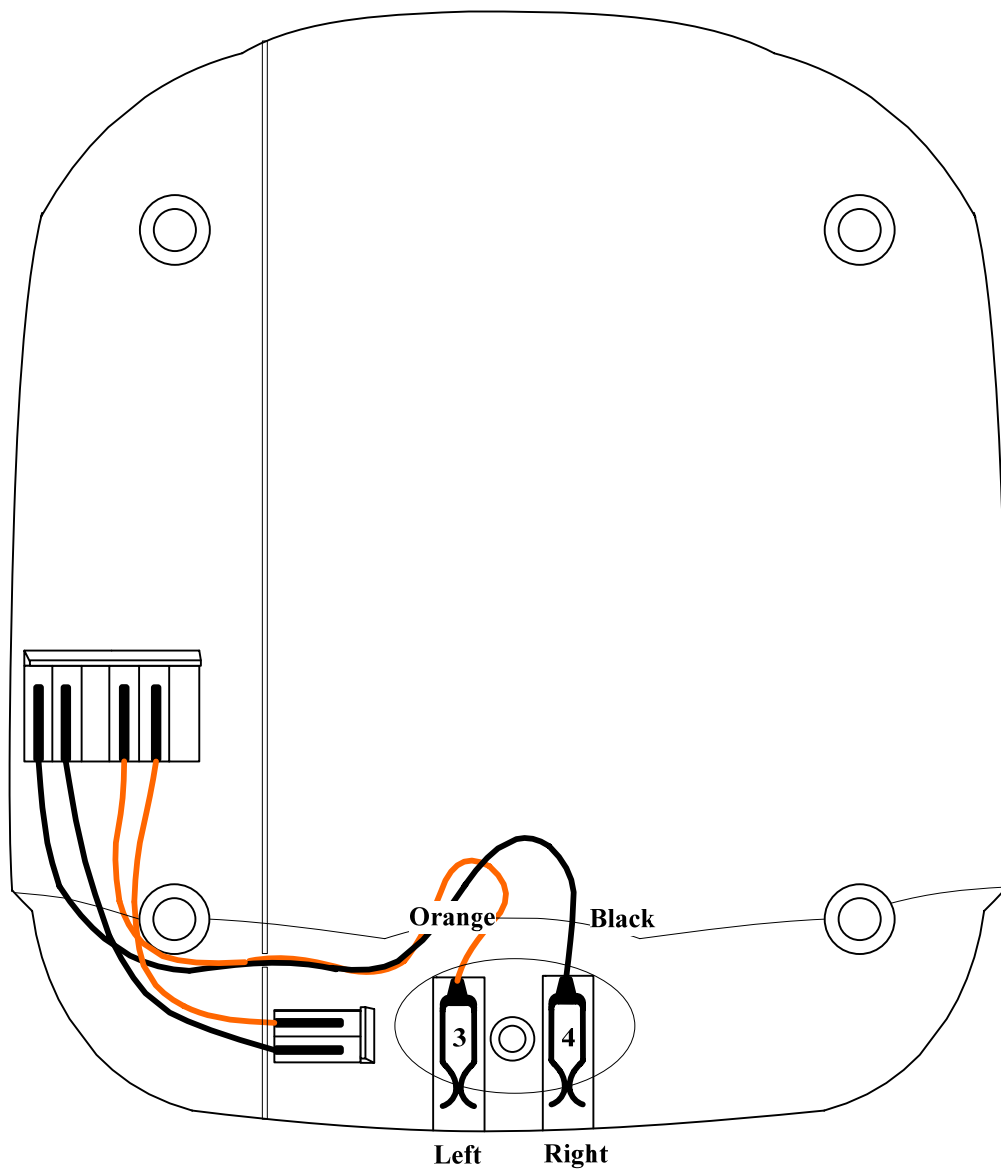


Fig. F – Power supply to humidifier

**Connecting relationship:**

| Fig. \ Part       | 1              | 2           | 3            | 4             |
|-------------------|----------------|-------------|--------------|---------------|
| D                 | Back of device | Fig. C: CN1 |              |               |
| E                 | Fig. C: CN2    | Fig. A: P24 | Fig. F: Left | Fig. F: Right |
| Blower 8-pin plug | Fig. A: PM     |             |              |               |

**Table 2**

## 2. Humidifier

Fig. G – Humidifier PCBA (back side)

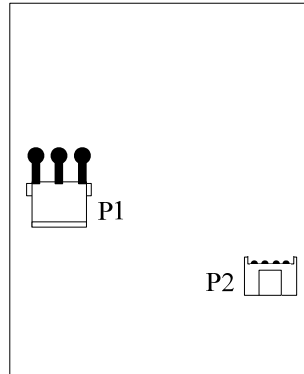


Fig. H – Power supply to humidifier

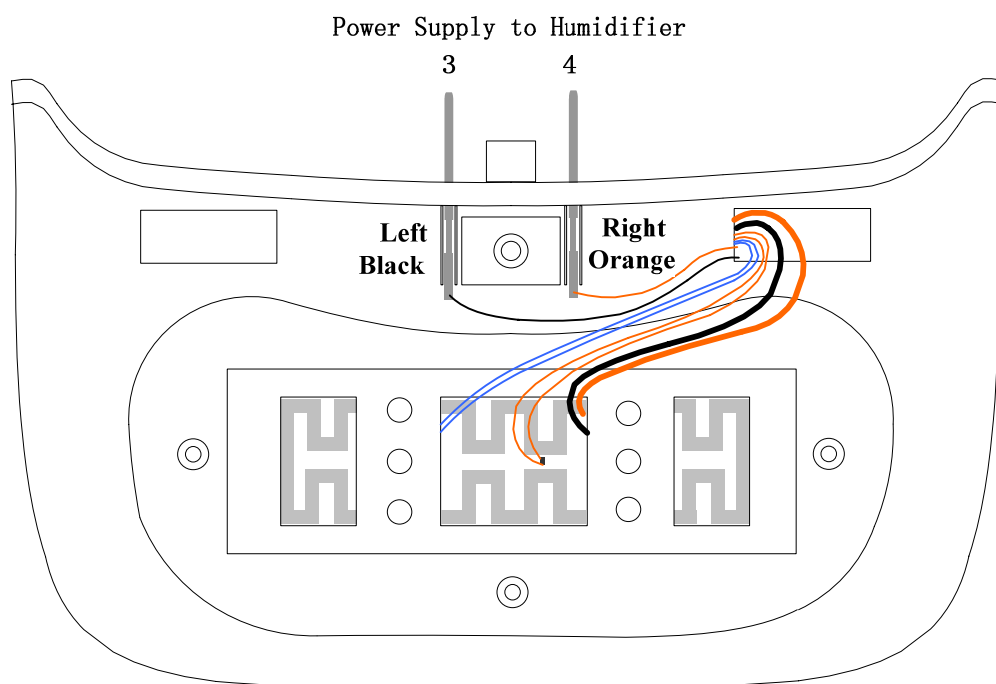
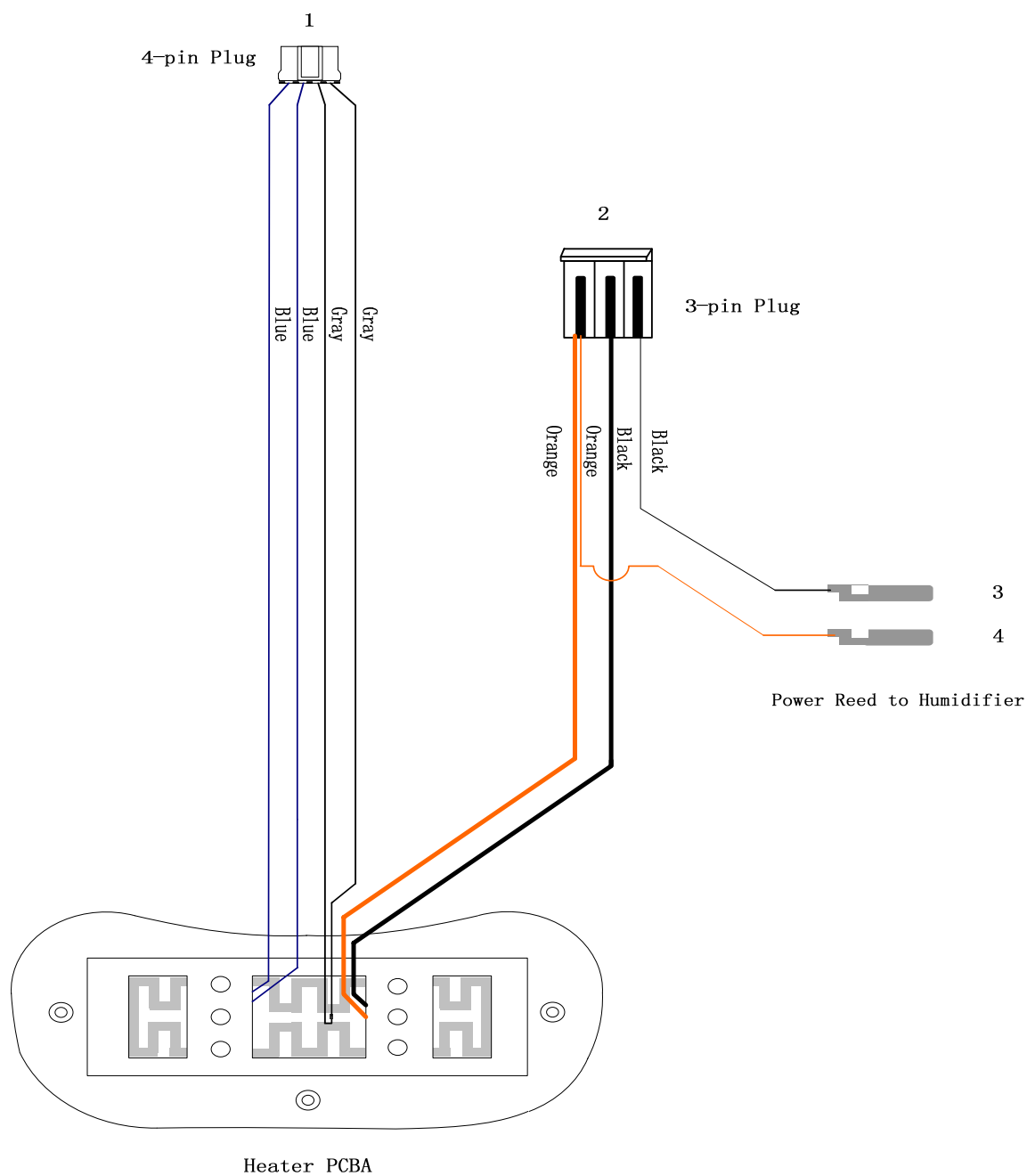


Fig. I – Heater PCBA



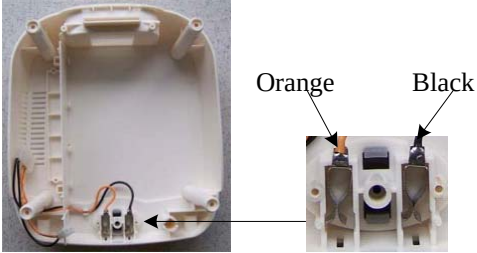
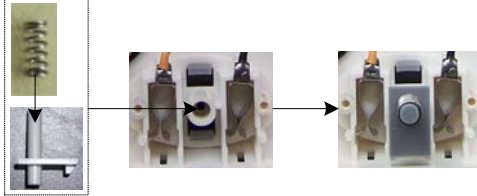
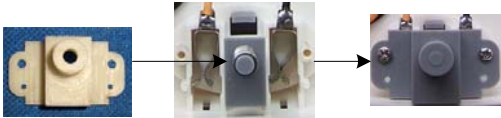
**Connecting relationship:**

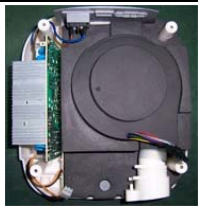
| Fig. \ Part | 1          | 2          | 3            | 4             |
|-------------|------------|------------|--------------|---------------|
| I           | Fig. G: P2 | Fig. G: P1 | Fig. H: Left | Fig. H: Right |

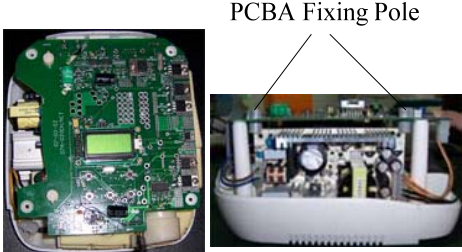
**Table 3**

## Components Illumination

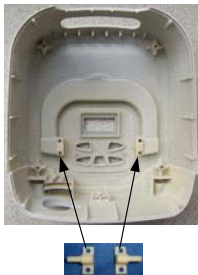
### 1. Device Inside

| S/N | Assembly                                                                      | Figure                                                                                |
|-----|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1   | Insert humidifier locker on the shield bottom.                                |     |
| 2   | Insert humidifier power supply reed on the shield bottom as per Fig. E and F. |     |
| 3   | Assembly the humidifier locker and button spring.                             |    |
| 4   | Put fixing on the spring and fix by screw (2× 3mm*8mm).                       |   |
| 5   | Put power wire.                                                               |   |
| 6   | Assembly the shield back on the bottom, fix by screw (2× 3mm*8mm).            |  |
| 7   | Assembly foam (bottom) into shield bottom.                                    |  |
| 8   | Assembly AC power socket on the shield back (character face up).              |  |

|    |                                                                                                               |                                                                                                                                     |
|----|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 9  | Clean up the wire.                                                                                            |                                                  |
| 10 | Assembly power PCBA in the socket and connect all plugs. Clean up the wire.                                   |                                                  |
| 11 | Connect outlet to blower connector, fix by clip.                                                              |                                                  |
| 12 | Connect blower to connector and fix by clip. Put the integrated one in the right place.                       |                                                 |
| 13 | Covered by foam cover and clean up wires.                                                                     |                                                |
| 14 | Assembly fixing clip on the main board PCBA.                                                                  |  <p>PCBA Fixing Clip</p> <p>Pressure Sensor</p> |
| 15 | Plug the 8-pin plug of blower in the socket PM plug #2 as in Fig. E in the socket P24 on the main board PCBA. |                                                |

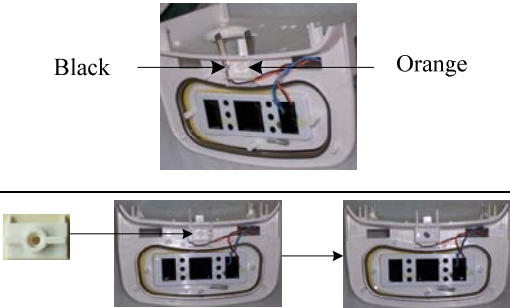
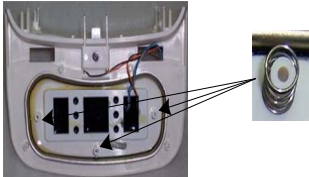
|    |                                                                                       |                                                                                    |
|----|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 16 | Assembly the main board PCBA on fixing pole and make sure the clip clamps power PCBA. |  |
|----|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|

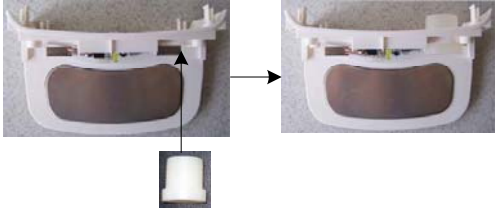
## 2. Device Outside

| S/N | Assembly                                                       | Figure                                                                                |
|-----|----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 18  | Put handle in the socket on shield cover.                      |   |
| 19  | Assembly handle fixing to handle and fix by screw (4x3mm*8mm). |  |
| 20  | Assembly the infrared window.                                  |   |
| 21  | Assembly and fix the panel.                                    |  |
| 23  | Put on the key button.                                         |  |

|    |                                                                                                                     |                                                                                                                                     |
|----|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 24 | Assembly shield cover on the bottom, adjust the position of outlet and make sure it is clamped in the right socket. |                                                                                                                                     |
| 25 | Fix the shield by screws (4×M3*10mm).                                                                               |  <p>Screw Hole</p>                                |
| 26 | Stick four device pads.                                                                                             |  <p>Device Pad</p>                                |
| 27 | Put in filter and assembly the cover.                                                                               |                                                  |
| 28 | Put on all plugs if necessary (without humidifier).                                                                 |  <p>Plug (up)<br/>Power plug<br/>Plug (down)</p> |

### 3. Humidifier

| S/N | Assembly                                                                        | Figure                                                                                                          |
|-----|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1   | As per Fig. H, assembly the heater plane and wire. Fix them by screw (3mm*8mm). |  <p>Black</p> <p>Orange</p> |
| 2   | Put spring on each of three poles on heater plane.                              |                            |

|   |                                                                         |                                                                                                                                                                                          |
|---|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | Assembly the scaleboard on the heater plane and fix by screw (3mm*8mm). |   <p>Scaleboard</p> |
| 4 | As per Table 3, assembly heater PCBA and wires.                         |                                                                                                       |
| 5 | Put the humidifier inner connector on the shield.                       |                                                                                                        |
| 6 | Put on light window.                                                    |                                                                                                       |
| 7 | Assembly the inner and outer shield.                                    |                                                                                                      |
| 8 | Fix on screw (M3*10mm).                                                 |  <p>Screw</p>                                                                                        |

**Appendix B:****Spare Part Configuration**

| S/N | Name              | Qty | Figure                                                                              |
|-----|-------------------|-----|-------------------------------------------------------------------------------------|
| 1   | Shield Cover      | 1   |    |
| 2   | Handle            | 1   |    |
| 3   | Handle Fixing     | 2   |   |
| 4   | Shield Bottom     | 1   |  |
| 5   | Humidifier Clip   | 1   |  |
| 6   | Humidifier Locker | 1   |  |
| 7   | Humidifier Fixing | 1   |  |
| 8   | Filter Cover      | 1   |  |
| 9   | Shield Back       | 1   |  |

|    |                             |   |                                                                                     |
|----|-----------------------------|---|-------------------------------------------------------------------------------------|
| 10 | Outlet                      | 1 |    |
| 11 | PCBA Fixing                 | 2 |    |
| 12 | Infrared Window             | 2 |    |
| 13 | Panel                       | 1 |    |
| 14 | Humidifier Outer Shield     | 1 |    |
| 15 | Humidifier Inner Shield     | 1 |  |
| 16 | Light Window                | 1 |  |
| 17 | Pole Platelet               | 1 |  |
| 18 | Humidifier Power Plug       | 1 |  |
| 19 | Humidifier Plug (left-up)   | 1 |  |
| 20 | Humidifier Plug (left-down) | 1 |                                                                                     |

|    |                                 |   |                                                                                     |
|----|---------------------------------|---|-------------------------------------------------------------------------------------|
| 21 | Humidifier Plug<br>(right-up)   | 1 |    |
| 22 | Humidifier Plug<br>(right-down) | 1 |                                                                                     |
| 23 | Device Pad                      | 6 |    |
| 24 | Blower Connector                | 1 |    |
| 25 | Humidifier Inner<br>Connector   | 1 |    |
| 26 | Key Button                      | 1 |  |
| 27 | Humidifier<br>Scaleboard        | 1 |  |
| 28 | Heater Plane Spring             | 3 |  |
| 29 | Button Spring                   | 1 |                                                                                     |

## Spare Part List

| Part Name                    | S/N         | Qty | Unit | Classify | Use         |
|------------------------------|-------------|-----|------|----------|-------------|
| Main Board PCBA              | 20211151    | 1   | PC   | PCBA     | Main Device |
| Heater PCBA                  | 20212-41    | 1   | PC   | PCBA     | Humidifier  |
| Power Supply PCBA            | 20311101    | 1   | PC   | PCBA     | Main Device |
| Blower                       | 40211       | 1   | PC   | Assembly | Main Device |
| Heater Plane                 | 45010101    | 1   | Set  | Assembly | Humidifier  |
| Power Supply PCBA Wire       | 45010201    | 1   | Set  | Wire     | Main Device |
| Main Board PCBA Wire         | 45010301    | 1   | Set  | Wire     | Main Device |
| Silicon Rubber Canal         | 50224       | 1   | PC   | Assembly | Main Device |
| Foam (bottom)                | 40191-0101  | 1   | PC   | Foam     | Main Device |
| Foam (cover)                 | 40191-0201  | 1   | PC   | Foam     | Main Device |
| Shield Cover                 | 4011110101  | 1   | PC   | Plastic  | Main Device |
| Handle                       | 40111130201 | 1   | PC   | Plastic  | Main Device |
| Handle Fixing                | 40111-0301  | 2   | PC   | Plastic  | Main Device |
| Shield Bottom                | 4011110401  | 1   | PC   | Plastic  | Main Device |
| Humidifier Clip              | 40111130501 | 1   | PC   | Plastic  | Main Device |
| Humidifier Locker            | 40111-0601  | 1   | PC   | Plastic  | Main Device |
| Humidifier Fixing            | 40111-0701  | 1   | PC   | Plastic  | Main Device |
| Filter Cover                 | 40111130801 | 1   | PC   | Plastic  | Main Device |
| Shield Back                  | 40111130901 | 1   | PC   | Plastic  | Main Device |
| Outlet                       | 4011123001  | 1   | Set  | Plastic  | Main Device |
| PCBA Fixing                  | 40111-1301  | 2   | PC   | Plastic  | Main Device |
| Infrared Window              | 4011101401  | 2   | PC   | Plastic  | Main Device |
| Panel                        | 4011210101  | 1   | PC   | Plastic  | Main Device |
| Power Supply PCBA Fixing     | 40111-1601  | 1   | PC   | Plastic  | Main Device |
| Humidifier Outer Shield      | 4011112201  | 1   | PC   | Plastic  | Humidifier  |
| Humidifier Inner Shield      | 4011112301  | 1   | PC   | Plastic  | Humidifier  |
| Pole Platelet                | 40111-2601  | 1   | PC   | Plastic  | Humidifier  |
| Light Window                 | 4011142401  | 1   | PC   | Plastic  | Humidifier  |
| Button Spring                | 4013140101  | 1   | PC   | Metal    | Main Device |
| Humidifier Scaleboard        | 4013130401  | 1   | PC   | Metal    | Humidifier  |
| Heater Plane Spring          | 4013140501  | 3   | PC   | Metal    | Humidifier  |
| Humidifier Power Plug        | 4012110101  | 1   | PC   | Rubber   | Main Device |
| Humidifier Plug (left-up)    | 4012110201  | 1   | PC   | Rubber   | Main Device |
| Humidifier Plug (left-down)  | 4012110301  | 1   | PC   | Rubber   | Main Device |
| Humidifier Plug (right-up)   | 4012110401  | 1   | PC   | Rubber   | Main Device |
| Humidifier Plug (right-down) | 4012110501  | 1   | PC   | Rubber   | Main Device |
| Device Pad                   | 4012110601  | 6   | PC   | Rubber   | All         |
| Key Button                   | 4012110701  | 1   | PC   | Rubber   | Main Device |
| Blower Connector             | 4012110801  | 1   | PC   | Rubber   | Main Device |

|                            |            |   |    |           |             |
|----------------------------|------------|---|----|-----------|-------------|
| Humidifier Inner Connector | 4012110901 | 1 | PC | Rubber    | Humidifier  |
| Screw-1                    |            | 8 | PC | Screw     | All         |
| Screw-2                    |            | 9 | PC | Screw     | All         |
| Sunk Screw                 |            | 4 | PC | Screw     | Main Device |
| Filter                     |            | 1 | PC | Accessory | All         |
| Power Cord                 | 33011101   | 1 | PC | Accessory | All         |
| Water Tank                 | 45010401   | 1 | PC | Accessory | Humidifier  |